

TRAINING & OPTIMIZATION		
Optimizer	Adam (identical)	
Batch size	512 (identical)	
Supervised epochs (main run)	10 (identical)	
Supervised epochs (secondary tiers)	5 (paper: 10)	
Dropout	0.0 (identical)	
Weight decay / L2	0.0 (identical)	
Random seed(s)	123, 124 (not in paper)	
LR sweep grid, supervised	{0.01, 0.001} (paper: +0.0001)	
LR sweep grid, unsupervised	{2e-6, 2e-7} (paper: +2e-8)	
Final LR, supervised	0.01 (all 4 aggregators) (not in paper)	
Final LR, unsupervised	2e-6 (m/s/pool), 2e-7 (gcn) (not in paper)	
Framework version	TensorFlow 1.15.0 (not in paper)	

PPI-SPECIFIC STUDIES ADDED BY THIS REPO	
K-sensitivity sweep	K ∈ {1, 2, 3}†, 5 epochs (not in paper)
Noise-robustness test	noise ∈ {0.0, 0.5, 1.0}, GCN & pool (not in paper)
Multi-seed variance check	seeds {123, 124}†, mean ± std F1 (not in paper)
Embedding snapshots tracked	steps {0,100,200,800,3k,8k,17,050} (not in paper)
Snapshot model / sample	graphsage_maxpool (unsup), 1,000 nodes (not in paper)
Embedding projection	PCA + t-SNE (perpl. 30), seed 123 (not in paper)
† K=2 and seed=123 reuse the original full 10-epoch run; every other point (K=1/K=3, seed=124) is freshly trained at the reduced 5-epoch budget — a mixed comparison, not a clean one (see README "Scaled local reproduction"). Excluded: Lemma 1/2/3 proof constants (C, ε, b), Appendix E — theoretical, not implementation parameters.	

AGGREGATOR ARCHITECTURES		
Aggregators run for PPI	mean, GCN, LSTM, max-pool (identical)	
Recursive embedding dim h_v^k	256 / layer, all 4 aggregators (identical)	
— how GCN reaches 256	layer dim passed as 2×dim_1/dim_2 (mechanism not in paper)	
Self neighbor concatenation	on; off for GCN (identical)	
Pooling MLP hidden dim ("small")	512 (identical)	
LSTM hidden dim ("small")	128 (identical)	
Weight initialization	Glorot / Xavier uniform (not in paper)	
Gradient clipping	[-5, 5] (not in paper)	

UNSUPERVISED OBJECTIVE		
Loss	skip-gram + negative sampling (identical)	
Negative samples Q	20 (identical)	
Negative-sampling distortion	0.75 (identical)	
Context-pair generation	50 walks × length 5 (identical)	
Embedding normalization	unit L2-norm, final embedding only (paper's Alg. 1/2 pseudocode: per-hop)	
Unsupervised steps (sweep)	3,000 (of ≈17,050/epoch) (not in paper)	
Live training diagnostic	Unsupervised MRR (not in paper)	

COMPUTE ENVIRONMENT		
GPU	1× RTX 2060 SUPER (paper: 4× Titan X Pascal, 16 Xeon CPUs)	
Total wall-clock, scaled run	≈75 min (PPI only, ~100 min budget) (paper: ≈3 days, all 3 datasets + DeepWalk)	

DATASET & TASK SETUP		
Graph split	20 / 2 / 2 (train/val/test) (identical)	
Total nodes	56,944 (not in paper)	
Total edges	818,716 (not in paper)	
Input feature dimensionality	50-dim (not in paper)	
Labels	121 GO terms, multi-label (identical)	
Feature sparsity	42% nodes zero-valued (identical)	
Loss head for PPI	sigmoid cross-entropy (identical)	

ALGORITHM DEPTH & NEIGHBOR SAMPLING		
Search depth K	2 (identical)	
Sample sizes S ₁ , S ₂	25, 10 (identical)	
Sampling method	uniform, with replacement (identical)	
Max-degree cap (supervised)	128 (identical)	
Max-degree cap (unsupervised)	100 (paper: 128)	

EVALUATION & BASELINES		
Final metric	micro-averaged F1 (identical)	
Baselines reproduced	Random, Raw features (identical)	
Baseline classifier	SGDClassifier, log loss, max_iter=5 (not in paper)	
DeepWalk / node2vec baseline	not executed for PPI (identical — paper excludes it too, App. D)	